

Information Sharing, Holdup, and External Finance: Evidence from Private Firms

Andrew Bird

Tepper School of Business, Carnegie Mellon University

Stephen A. Karolyi

Tepper School of Business, Carnegie Mellon University

Thomas G. Ruchti

Tepper School of Business, Carnegie Mellon University

To mitigate holdup by an informed incumbent lender, a private borrower may publicly share information in order to increase lender competition. Despite proprietary costs, a subset of private borrowers voluntarily share private information in loan and credit underwriting agreements. These borrowers switch lenders at a 16% higher rate and receive lower loan financing costs. For private firms that go public, we analyze changes in the net benefits of information sharing and study the potential estimation bias from unobservable borrower quality. This setting corroborates our inference that voluntary information sharing reduces lender holdup and alleviates financial constraints for private firms. (*JEL* G21, G28, G32, M41)

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Relationships are an important feature of the syndicated loan market.¹ A consequence of these relationships is that banks acquire private information about their borrowers (Hauswald and Marquez 2006). This information asymmetry between the relationship lender and potential outside lenders makes it costly for borrowers to switch lenders, since outside lenders face a winner's curse in competing to lend to the borrower. These switching costs allow the relationship lender to hold up the borrower by raising spreads (Schenone 2010; Bharath et al. 2007). In this paper, we show that borrowers respond to

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¹ See, for example, Petersen and Rajan (1994), Berger and Udell (1995), Chava and Purnanandam (2011), Gopalan, Udell, and Yerramilli (2011), Chodorow-Reich (2013), Carvalho, Ferreira, and Matos (2015), and Schwert (2018).

this asymmetry by publicly sharing information, which, by increasing lender competition, reduces spreads and increases access to credit (Sharpe 1990; Rajan 1992; Berger and Udell 1995, Petersen and Rajan 1994).²

We measure borrower information sharing using voluntary disclosures of financial measures included in loan documentation collected by the largest loan agreement aggregator and disseminator (Loan Pricing Corporation [LPC]). This is an ideal setting in which to study the economics of borrower information sharing, since we can observe both the mechanism—an increase in lender competition—through lender switching, and the effects, through loan contracting terms and financing outcomes. Borrower information sharing through LPC may be valuable to borrowers because all potential lenders have access to the data. In fact, according to LPC, their information is “used every day by banks, asset managers, law firms, regulators, corporations and others to drive valuation, syndication, trading, and research and portfolio management activities” (Thomson Reuters 2018). Similarly, these disclosures can inform product market competitors either directly through their own access to the data or indirectly through their own lending relationships. Despite such proprietary costs, 6% of private borrowers voluntarily share information.

We find that information sharing by borrowers increases competition by lenders. For subsequent loans, disclosing borrowers are 16% more likely to switch lenders and borrow from larger syndicates, where the retained share of the lead bank is smaller, and with greater institutional participation. These findings are consistent with information sharing acting to reduce the information disadvantage of potential alternative lenders, spurring competition to participate in these loans. Not surprisingly, this reduction in lender holdup and improvement in borrower bargaining power manifests itself in an improvement in financing outcomes. Disclosing borrowers pay 4% lower spreads and receive 8% higher loan amounts, suggesting that borrower information sharing can play an important role in alleviating financial constraints for private firms.

These significant benefits of information sharing raise a natural question – why do only a minority of borrowers choose to share information? One possibility is that borrowers may have alternative, potentially less costly, ways to mitigate the holdup problem, such as privately sharing information with particular outside lenders.³ Another possibility is borrower heterogeneity, which we address by including in our preferred empirical specification a number of loan characteristics as control variables and borrower and lender fixed effects. The latter strategy eliminates potentially confounding, time-invariant

² Petersen and Rajan (1994) summarizes this theoretical literature, and their work is suggestive of the mechanism we study in the context of competition determining the extent of holdup. They note that “the state of competition depends, of course, on the number of potential lenders in the market and on *how informed they are*” (p. 6, emphasis added).

³ However, public disclosure and private disclosure also could be complements, with public disclosure reducing screening costs and leading to negotiations involving the private sharing of much more information. We do not pursue this point because of a lack of data on private information sharing.

characteristics while the former strategy mitigates the confounding effects of time-varying borrower attributes as loan terms incorporate lenders' private information that is unobservable to the econometrician. However, a significant amount of cross-sectional variation in information sharing remains, implying that borrowers must incur some countervailing costs of disclosure, which could come in many forms. Once firms make these financials a part of loan agreements, they can be litigated on, though this possibility could help convince outside lenders of the credibility of the disclosure, and so reinforce the benefits of information sharing. Second, the shared information could hurt borrowers by aiding product market competitors (Bhattacharya and Ritter 1983; Verrecchia 1983; Darrough and Stoughton 1990). Exploiting cross-sectional measures of proprietary costs, such as the level of competition, we find suggestive evidence that this factor significantly reduces the likelihood of borrower information sharing. Our evidence on the role of proprietary costs corroborates evidence from other voluntary disclosure settings, but is novel in the corporate loan market context.

The borrower's trade-off hints at the main identification concern with our empirical results: disclosing borrowers could be unobservably different from nondisclosing borrowers. Indeed, the theoretical intuition of Verrecchia (1983) provides the foundation for a plausible alternative explanation for our findings, since higher-quality borrowers are more likely to disclose to avoid pooling with nondisclosing borrowers. However, several factors push in the opposite direction. More successful firms face higher proprietary costs of disclosure in addition to a potentially higher benefit. Further, if lenders are relatively more interested in lending to higher-quality firms,⁴ then the marginal effect of disclosing on lender attention, and so lender competition, could be higher for lower-quality firms. Thus, the direction of any bias from unobservable borrower quality is theoretically ambiguous, and so must be investigated empirically.

The subsample of firms that eventually do an initial public offering (IPO) provides an ideal laboratory to study the trade-off underlying the borrower information sharing decision and better understand estimation bias due to unobserved borrower quality. As the firm gets close to the IPO, with all its attendant mandatory disclosure, the cost of disclosure falls, as the revelation of proprietary information to competitors becomes inevitable. Our data confirm this hypothesis: borrowers are significantly more likely to disclose the closer they get to their IPO. Following the instrumental variables (IV) approach of Altonji and Shakotko (1987),⁵ this firm-specific time series variation can be

⁴ This comparative static hold, even when spreads adjust to borrower quality, if there exists a borrower quality threshold, below which a mutually beneficial loan contract does not exist.

⁵ Altonji and Shakotko (1987) study the effect of tenure on wages. Their empirical concern is that tenure is positively correlated with the unobserved quality of the employee-employer match, and so a simple OLS regression would yield upward-biased estimates of the effect of tenure on wages because it captures the fact that better matching

used to deal with unobserved borrower quality. Specifically, the instrument is the firm-specific proportion of time left to IPO or the time to IPO, divided by the number of years a firm remains private. This strategy collapses time to IPO into the same unit interval for all firms, effectively creating a quality-adjusted disclosure trend for each firm. Our IV estimates are typically larger in magnitude than the OLS estimates described above, suggesting that, if anything, lower-quality borrowers are more likely to disclose.

We also take an alternative identification approach of proxying for fixed firm quality using the number of years from initial appearance in the loan sample to the eventual IPO, following the logic that, on average, higher-quality firms are faster to IPO, and by proxying for time-varying firm quality using the time left until the IPO. Interestingly, even though the identifying variation in borrower information sharing under this approach is very different from that used by the IV approach, we obtain similar results. We believe that this multipronged empirical strategy, including the use of borrower and lender fixed effects in our main tests, provides strong evidence that it is indeed information sharing by the borrower, rather than unobservable differences, that explains the increase in lender competition and improvement in financing outcomes.

The most closely related paper to ours is Schenone (2010), who uses IPOs as a shock to the information environment of borrowers and finds that lenders use their information advantage over borrowers to extract rents when negotiating loan terms. Building on Detragiache, Garella, and Guiso (2000), Schenone (2010) also provides evidence that borrowing from multiple banks at the same time can mitigate holdup problems, which suggests that private borrowers might strategically borrow from multiple sources to reduce the information monopoly from any one source.⁶ Our findings suggest that private borrowers strategically respond to this same holdup problem by publicly sharing information to level the playing field for potential alternative lenders.⁷ Because potential lenders may be not only uninformed about the prospects of many private firms but also unaware of their existence, voluntary information sharing may benefit private borrowers along both margins in ways that are inaccessible to other types of firms.

The literature on *mandatory* information sharing in credit markets through credit bureaus and registries elucidates a mechanism similar to that used in our

leads to longer tenure. The authors deal with this problem by instrumenting for tenure with a variable representing the fraction of the total tenure elapsed at any given time. As argued in Altonji and Shakotko (1987), this instrument eliminates time-varying unobservable quality by placing all employee tenures on the same scale. This approach also has been used in the finance and accounting literature to examine the interaction of earnings management and tenure on CEO compensation (Cella, Ellul, and Gupta 2017).

⁶ Houston and James (1996) provide evidence of a related benefit of a reduction in the lender's monopoly power: an increase in investment efficiency (Rajan 1992; Diamond 1993).

⁷ These strategies are not mutually exclusive, and borrowers' equilibrium selections of holdup busting actions are likely to vary with the costs and benefits of each strategy. On margin, for example, relative to public information sharing, contracting with multiple banks simultaneously incurs not only lower proprietary costs but also larger direct transaction costs.

paper (Simon 1989; Musto 2004; Bushee and Leuz 2005; Greenstone, Oyer, and Vissing-Jorgensen 2006; Qian and Strahan 2007; Hertzberg, Liberti, and Paravisini 2010). While the net benefits of mandatory disclosure regulation may be positive, mandatory disclosure does not perfectly substitute for voluntary disclosure, and, in particular, may compromise voluntary disclosure as a means for borrowers to efficiently match with banks.

Another literature in finance and accounting studies managerial incentives to alter information flows in anticipation of financing activities. Verrecchia and Weber (2006) examine redacted disclosure in material contracts filed with the SEC and show that firms are less likely to redact information when they are planning to raise capital. Illustrating the same mechanism but with reversed managerial incentives, Brockman, Khurana, and Martin (2008) find that firms opportunistically make more frequent bad news announcements prior to repurchasing shares, while Darrough and Rangan (2005) illustrate how managers employ both real and accruals-based earnings management prior to IPO. Further, Jo and Kim (2007) show that a higher frequency of disclosure leads to better post-seasoned equity offering performance. Khurana, Pereira, and Martin (2006) find a positive relationship between firm disclosure policy and the externally financed growth rate, consistent with our finding that disclosure can relax financial constraints for private firms. Similarly, Lang and Lundholm (2000) presents evidence consistent with voluntary disclosure around seasoned equity offerings (SEOs) being used to “hype” the stock—prior to the offering, returns are abnormally high but are actually negative surrounding the announcement of the SEO. Given that the firms in our sample are not publicly traded, this particular mechanism is unlikely to be at play, though it is related to our idea of advertising to potential lenders.

More generally, our paper contributes to the literature on the determinants and consequences of voluntary disclosures (Healy and Palepu 2001). A subset of this literature documents a link between increased disclosure (through, for example, managerial forecasts) and a lower cost of capital (e.g. Botosan 1997, Botosan and Plumlee 2002). The typical mechanism is that information sharing lowers risk. While we also study a benefit to voluntary disclosure, our mechanism is significantly different, occurring through an increase in lender competition. We provide empirical support for this less studied rationale for voluntary disclosure. Our investigation of the information sharing incentives of private firms also extends beyond the typical public company setting of this literature, which is particularly useful given that the dearth of mandatory disclosure from these firms should increase the impact and import of their voluntary disclosure (Ball, Jayaraman, and Shivakumar 2012; Einhorn 2005). We also contribute to the literature on IPO outcomes (Ritter and Welch 2002; Loughran and Ritter 2004; Ljungqvist 2005; Hanley and Hoberg 2010; Leone, Rock, and Willenborg 2007) by showing that a firm’s voluntary disclosure before an IPO can lead to better IPO outcomes.

1. Data and Institutional Background

1.1 Institutional background

Using Loan Pricing Corporation's (LPC) Dealscan database, we collect borrower, lender, and loan characteristics and match borrowers to lenders at the loan level to identify bank switching. Importantly, we also collect data on financial disclosures made in syndicated loans from the LPC Dealscan database. Among the financial ratios and amounts potentially disclosed in loan agreements and coded in our data, five are performance measures, including Sales, Debt/EBITDA, Senior Debt/EBITDA, Interest Coverage and Fixed Charge Coverage, and two are capital ratios: Leverage and Senior Leverage.⁸

The financial disclosures contained in Dealscan are the same ones that LPC provides to its institutional clients, who are active participants in the syndicated loan market. LPC is the primary supplier of global private loan market news, data, and analysis via print publications, online news, analysis, valuation services, and interactive databases. LPC's data products are available in real-time to banks, and are typically used by loan syndication divisions; more recently, they have been used by other financial institutions, including hedge funds, that participate in the private loan market, and financially sophisticated borrowers (Ivashina, Iverson, and Smith 2016; Fang, Ivashina, and Lerner 2013; Becker and Ivashina 2015; Ivashina and Kovner 2011; Benmelech, Dlugosz, and Ivashina 2012; Ivashina and Sun 2011). LPC's Dealscan database, which is used by academics and private corporations, contains historical data on over 220,000 loan and bond transactions. About 60% of the DealScan data come from SEC filings (Chava and Roberts 2008), including 10-Ks, 10-Qs, 8-Ks, and registration statements, but for loans to private borrowers, LPC also collects data directly from market participants, including borrowers and lenders.

The interpretation of empirical results in this paper depends on the fact that it is the borrower who makes an active disclosure decision, rather than the lender or the data provider. To verify that this is indeed the case, we have conducted extensive interviews with borrowers, lenders and LPC itself. From this process, we understand the data to be constructed as follows. First, the borrower decides whether to include additional information, such as sales, in the credit underwriting agreement. LPC acquires the data through two channels. Lenders share these underwriting agreements en masse with LPC, and LPC also contacts borrowers to supplement and check their data. This sharing by lenders is nonstrategic – the social norm in the industry is that all lenders share their agreements so that all players have access to unvarnished data, the most important use of which is in benchmarking loan terms. Importantly, the loan

⁸ Our findings are qualitatively robust to excluding sales disclosures.

officers we talked to not only denied strategically disclosing themselves but also stated their belief that their competitors are also not strategically disclosing.⁹

1.2 Data

Because of data availability constraints before 1995, we restrict our sample to loans initiated between 1995 and 2012. Our main sample contains 30,628 loan facilities issued by private borrowers. In Section 3, we investigate the subsample of loans of private borrowers who eventually undertake an IPO. Data on IPOs come from the Kenney-Patton IPO database (Kenney and Patton 2013). This sample contains 1,683 loan facilities of private borrowers that issue an IPO between 1995 and 2012. These loan facilities are representative of the full sample.

Panel A of Table 1 presents descriptive statistics about these performance and financial disclosures for our main sample. We measure *Disclose*, an indicator that equals 1 if at least one financial metric is disclosed and 0 otherwise, by identifying loans that (1) include one of seven metrics (i.e., Sales, Debt/EBITDA, Senior Debt/EBITDA, Leverage, Senior Leverage, Fixed Charge Coverage, Interest Coverage), (2) reference format or date of the borrower's financial statements, and (3) do not have a financial covenant written on the corresponding metric. In 5.8% of loans, borrowers make at least one such financial disclosure. Sales and Senior Debt/EBITDA are the most frequently disclosed metrics; about 60% of disclosures include at least one of these metrics. Senior Leverage is the least frequently disclosed metric; 11.0% of loans with a disclosure include this metric.

Panel B of Table 1 presents summary statistics about the borrowers, lenders, and loans in the sample. The average loan has an all-in-drawn spread of 284 bps, maturity of 50 months, and amount of \$107M. Most of these loans are arranged by lenders that have a prior relationship with the borrower as only 20.7% of loans are arranged by new lenders. However, 91.2% of loans are syndicated, and the average syndicated loan has three participants. Among syndicated loans, the lead arranger retains 36.5% of the loan, on average, a slightly higher percentage than the retained share for public borrowers (Sufi 2007), and the average loan has less than 0.01 nonbank institutions as participants (Fang, Ivashina, and Lerner 2013; Ivashina and Sun 2011).

Panel B of Table 1 also presents subsample means for private borrowers that make voluntary financial disclosures and those that do not. These univariate differences suggest that disclosers are less likely to borrow from sole lenders and more likely to subsequently initiate loans with new lenders. Disclosers subsequently issue loans to syndicates with a larger number of lenders, in which the lead arranger retains a smaller fraction of the loan, and nonbank institutions are participate at a higher rate. All of these univariate tests are consistent with

⁹ In Section 3, we show that our main results are robust to the inclusion of lender fixed effects, which is consistent with lenders playing a passive role in the construction of LPC's data.

Table 1
Summary statistics

A. Voluntary disclosures in loan agreements							
		Mean	SD	P25	Median	P75	
Disclose	5.75%						
Conditional on disclosure:							
Sales	58.82%	1,570	5,090	185	479	1,290	
Debt/EBITDA	26.01%	3.71	20.86	3.09	4.26	5.54	
Senior Debt/EBITDA	62.40%	3.46	1.52	2.60	3.40	4.20	
Leverage	29.27%	0.90	1.74	0.45	0.68	0.89	
Senior leverage	11.03%	0.70	2.06	0.35	0.56	0.82	
Fixed charge coverage	19.11%	1.81	2.78	1.09	1.46	1.91	
Interest coverage	40.30%	3.71	5.75	1.74	2.33	3.66	
B. Summary statistics							
	Disclose	Not disclose	Full sample				
	Mean	Mean	Mean	SD	P25	Median	P75
Loan terms:							
Spread	343	282	284	157	175	275	350
Facility amt. (\$mm)	146	106	107	251	19	45	100
Maturity (months)	68	49	49.81	31.02	24	50	61
Num. of covenants	2.59	2.65	2.64	1.22	2	3	3
Syndicate structure:							
Sole lender (%)	4.72	8.96	8.84	—	—	—	—
New lender (%)	25.54	20.56	20.70	—	—	—	—
Number of lenders	3.67	3.02	3.04	3.62	1	2	3
Retained share (%)	31.69	36.72	36.52	25.54	16	30.77	50
Number of institutions	0.03	0.01	0.01	0.11	0	0	0
Disclosure costs:							
RandDIntensity (%)	6.63	12.58	9.29	31.71	0.73	1.64	10.94
HHI	0.48	0.52	0.49	0.30	0.24	0.42	0.75
NumCompetitors	22	20	21	39	3	8	22
MarginPersistence	0.17	0.21	0.19	0.19	0.05	0.15	0.37
ProfitMargin (%)	−0.05	−0.07	−0.06	20.98	−0.08	0.01	0.04
AverageGrowth (%)	0.20	0.17	0.19	0.27	0.04	0.1	0.27%
GrowthDispersion	1.40	1.22	1.32	4.08	0.15	0.33	0.73
LitigationRisk (%)	30.28	35.48	32.61	—	—	—	—

This table provides loan package-level summary statistics for private borrowers between 1995 and 2012. Panel A presents summary statistics for each of the financial ratios and amounts disclosed in loan agreements. Panel B presents loan, lender, and disclosure cost characteristics associated with loan packages that include financial disclosures and those that do not. Section 1 defines the variables.

disclosure mitigating lender holdup, which encourages lender competition for disclosers’ future deals.

On average, disclosers’ loans have 61 bps higher spreads, \$40 million larger loan amounts, 19 months longer maturity, and 0.06 fewer covenants. These statistics do not deliver a clear message concerning the existing terms of disclosers’ loans. That disclosers’ loans have fewer covenants is encouraging, since one may be concerned about a link between the number of covenants and the financial disclosures we measure. However, the link between the number of covenants and financial disclosures is unlikely to cause an inflationary bias in our estimates of the external financing effects of voluntary financial disclosures. This is because covenants are used when borrowers have low bargaining power and poor credit quality, and so are typically associated

with negative consequences (Freudenberg, Imbierowicz, Saunders, and Steffen 2017; Murfin 2012; Chava and Roberts 2008; Roberts 2015; Chava, Fang, and Prabhat 2014). Nonetheless, we control for the number of covenants and other contemporaneous loan terms in our multivariate regressions to further mitigate such concerns.

2. Information Sharing and Lender Holdup

In this section, we investigate why borrowers would publicly share information as part of loan documents. Because this sharing reflects a choice by the borrower, there must be some benefit to doing so. Our hypothesized mechanism follows from the canonical contracting models of Sharpe (1990), Rajan (1992), and Von Thadden (2004), who all feature lender holdup. The idea is that a lender learns private information about a firm through the process of lending and can then condition loan offers on this information. Since this information is unavailable to outside lenders, they face a potential winner's curse in competing to lend to the borrower because they will be more likely to make the winning loan offer when the inside lender's information is bad. This information asymmetry between inside and outside lenders endogenously creates a switching cost that provides scope for the inside lender to hold up the borrower. According to the theory, by reducing information asymmetry with outside lenders, public disclosure of key financial ratios should improve the borrower's outside options, which mitigates holdup and leads to better loan terms. In the following subsections, we investigate this mechanism using a simple regression framework:

$$Y_{is} = \theta \text{Disclose}_{it} + \rho X_{it} + \omega_i + u_j + v_t + \xi_{it}, \quad (1)$$

where Y_{is} is an outcome for firm i on its next loan made in some future time period $s > t$, Disclose_{it} is a dummy variable equal to 1 if the firm disclosed in year t , X_{it} is a set of loan controls derived from the loan agreement, ω_i are borrower fixed effects, u_j are lender fixed effects, and v_t represents year fixed effects. We cluster standard errors at the borrower level.¹⁰ The included loan controls are *SoleLender*, an indicator that identifies loans with a single lender, the money terms of the loan, including $\ln \text{Spread}$, $\ln \text{FacilityAmount}$, and *Maturity*, and *NumCovenants*, the number of financial covenants in the loan package.

This simple OLS strategy tests the relationship between voluntary loan disclosures by private firms and subsequent outcomes, like lender switching and future loan terms, controlling for firm characteristics measurable through the loan agreement, and changes in broad market conditions eliminated with

¹⁰ In these specifications, double clustering by borrower and year has limited incremental effect on standard error estimates, suggesting that borrower effects are large and that our preferred specification is effectively controlling for unobserved variation.

year fixed effects. However, a concern with this approach, especially given the paucity of firm-level characteristics available for private firms, is unobservable borrower quality, which could be defined in several ways.¹¹ A number of plausible theoretical mechanisms could tie together not only quality and disclosure but also quality and borrower outcomes, so providing an alternative explanation for our findings. For the results in this section, we rely on the controls described above, and especially the borrower fixed effects to deal with this issue. It is important to note that controlling for loan terms is particularly powerful in this setting. Since loan terms incorporate the lender's private information about the borrower, controlling for loan terms naturally mitigates the confounding effects of a host of characteristics that are observable to the lender but unobservable to the econometrician. Building on this approach, in Section 3, we discuss the theoretical trade-off underlying the borrower's decision to share information in order to better understand the implications of unobservable borrower quality for bias in our empirical results. Additionally, we propose an IV strategy to further investigate the direction of this bias.

2.1 Lender switching

The first possible outcome of borrower information sharing that we study is the incidence of switching lenders, as this relates directly to lender competition. While the theoretical literature typically does not highlight the lender switching decision per se, Rajan (1992), for example, implies that reducing the information disadvantage of outside lenders increases the likelihood of bids by outside lenders, which increases the likelihood of switching. More generally, one would expect an increase in switching following a shock to the borrower's outside option in a model with imperfect bargaining. The borrower attracts more outside attention via public disclosure,¹² and can then threaten to leave unless the inside lender offers improved terms. We directly test this mechanism by studying the relationship between voluntary information sharing and lender switching.

We do so by using a left-hand side variable that is a dummy equal to 1 if the firm uses a different lender in its next loan. In the case of a loan syndicate, the dummy variable is equal to 1 if the lead arranger is different. In these tests, we restrict the sample to ensure that the lead bank can be identified in the current and subsequent loans, yielding a sample of 8,154 loans. Table 2 presents the results of estimating Equation (1). Column 1 shows the coefficient for the simple OLS model, which is positive and significant. If a firm shares information as part of a loan, then it is 7.0 percentage points more likely to switch lenders

¹¹ For our purposes, the relevant definition of borrower quality is any borrower attribute or type that would lead to better external financing outcomes.

¹² Voluntary information sharing may increase lender competition through a number of channels: by making potential lenders aware of the borrower, by increasing the precision of their information about the borrower, or by reducing screening costs.

Table 2
Voluntary disclosure and future lender choice

	<i>NewLender_{is}</i>			
	(1)	(2)	(3)	(4)
<i>Disclose_{it}</i>	7.0540* (4.1406)	7.5597** (3.8634)	14.1478*** (3.0371)	16.0735*** (3.2262)
Loan controls	No	No	No	Yes
Fixed effects:				
Year	Yes	Yes	Yes	Yes
Borrower	No	Yes	Yes	Yes
Lender	No	No	Yes	Yes
<i>R</i> ²	.0339	.5003	.5434	.5462
Obs.	8,154			

This table presents fixed effects regression estimates of *NewLender_{is}*, an indicator that equals 1 if borrower *i* switches lenders on its next loan (made in period *s* > *t*) and 0 otherwise, on *Disclose_{it}*, an indicator that equals 1 if firm *i* discloses at least one financial ratio or amount in its year *t* loan and 0 otherwise, and controls. We include borrower and lender fixed effects to isolate variation in the voluntary disclosure choice that varies across borrowers for the same lender and across loans for the same borrower, and we include year fixed effects to eliminate time-varying economic conditions that might affect voluntary disclosure choices and lender switching behavior. Controls include *SoleLender*, an indicator that identifies loans with a single lender, the money terms of the loan, including *lnSpread*, *lnFacilityAmount*, and *Maturity*, and *NumCovenants*, the number of financial covenants in the loan package. Robust standard errors are clustered by borrower and reported in parentheses. * *p* < .1; ** *p* < .05; *** *p* < .01.

for a future loan. Given that the likelihood of switching lenders in the sample is 20.7%, the effect of disclosure is to significantly increase the probability of future lender switching, consistent with the theoretical predictions discussed above.

In Columns 2 through 4, we successively add borrower and lender fixed effects and loan terms as control variables. Together, they somewhat amplify the effect of disclosure on lender switching to 16.1 percentage points. The change in magnitude is particularly due to the cross-sectional fixed effects, suggesting that persistent borrower and lender characteristics are associated with both the likelihood of sharing information and the incidence of lender switching. Overall, the results in this table are consistent with the theoretical intuition that switching costs are reduced as public information sharing by the borrower helps to level the playing field between potential new lenders and the incumbent lender (Rajan 1992). We come back to the additional possibility that the marginal benefit of switching also might increase with disclosure in Section 2.3.

2.2 Syndicate structure

Because lending is commonly done by syndicates rather than individual lenders, we further test the lender competition channel of voluntary information sharing by investigating the syndicate structure of subsequent loans. In this setting, enhanced lender competition should be reflected in greater demand to be a part of the syndicate. Thus, in Table 3, we investigate the effects of borrower information sharing on changes in the number of lenders in the syndicate (panel A), the retained share of the lead arranger (panel B), and the number of nonbank institutions participating in the loan (panel C). The sample size

Table 3
Voluntary disclosure and future syndicate structure

A. Syndicate size				
	(1)	(2)	(3)	(4)
<i>Disclose_{it}</i>	0.1942*** (0.0576)	0.2586*** (0.0885)	0.2711*** (0.0916)	0.2461** (0.1021)
Loan controls	No	No	No	Yes
Fixed effects:				
Year	Yes	Yes	Yes	Yes
Borrower	No	Yes	Yes	Yes
Lender	No	No	Yes	Yes
<i>R</i> ²	.0046	.3032	.3291	.3329
Obs.	14,273			
B. Retained share				
	(1)	(2)	(3)	(4)
<i>Disclose_{it}</i>	-8.8448** (3.2724)	-7.8381* (4.2922)	-5.4555 (3.5576)	-5.4755** (2.7914)
Loan controls	No	No	No	Yes
Fixed effects:				
Year	Yes	Yes	Yes	Yes
Borrower	No	Yes	Yes	Yes
Lender	No	No	Yes	Yes
<i>R</i> ²	.6325	.6506	.6956	.7227
Obs.	1,868			
C. Nonbank institutions				
	(1)	(2)	(3)	(4)
<i>Disclose_{it}</i>	0.0531** (0.0203)	0.0482** (0.0197)	0.0478** (0.0197)	0.0366** (0.0179)
Loan controls	No	No	No	Yes
Fixed effects:				
Year	Yes	Yes	Yes	Yes
Borrower	No	Yes	Yes	Yes
Lender	No	No	Yes	Yes
<i>R</i> ²	.0110	.0444	.0566	.0630
Obs.	26,442			

This table presents fixed effects regression estimates of *NumLenders_{is}*, the difference between the number of lenders participating in borrower *i*'s next and current loans (made in period $s > t$), *RetainedShare_{is}*, the difference between the lead bank's retained share on borrower *i*'s next and current loans, and *NumInstitutions_{is}*, the difference between the number of nonbank institutions participating in borrower *i*'s next and current loans at initiation, on *Disclose_{it}*, an indicator that equals 1 if firm *i* discloses at least one financial ratio or amount in its year *t* loan and 0 otherwise, and controls. We include borrower and lender fixed effects to isolate variation in the voluntary disclosure choice that varies across borrowers for the same lender and across loans for the same borrower, and we include year fixed effects to eliminate time-varying economic conditions that might affect voluntary disclosure choices and syndicate structure. Controls include *SoleLender*, an indicator that identifies loans with a single lender, the money terms of the loan, including *LnSpread*, *LnFacilityAmount*, and *Maturity*, and *NumCovenants*, the number of financial covenants in the loan package. Robust standard errors are clustered by borrower and reported in parentheses. **p* < .1; ***p* < .05; ****p* < .01.

varies across panels in Table 3 because of restrictions on observing current and future syndicate size, retained share, and participant bank identifiers. The most restrictive of these is retained share, and the sample in panel B falls to 1,868 loans since bank allocations are infrequently observed for private borrowers. In panel A, Column 1 shows that the number of lenders in future loans to disclosers is 0.2 higher, which corresponds to a 7% increase relative to the sample mean of 2.8 syndicate participants (including sole lender loans). That more lenders

participate in the loan is consistent with more demand and more competition for subsequent loans, as the disclosure reduces asymmetric information with other potential lenders. Columns 2 through 4 successively add borrower and lender fixed effects and loan terms as control variables and show similar results, such that fixed lender or borrower characteristics do not appear to explain the relationship between borrower information sharing and larger syndicates.

In panel B, the dependent variable is the change in the share of the loan retained by the lead arranger. Consistent with the above results, we expect that, all else equal, greater demand to be a part of the loan by other potential lenders would lead to a reduction in retained share, given the lead arranger's preference for diversification. Moreover, public information sharing by the borrower should reduce the asymmetric information problem between the lead and the syndicate participants, which is the typical rationale for the lead to hold part of the loan in the first place (Gorton and Pennacchi 1995; Sufi 2007). We find that borrower information sharing is associated with a decline in the retained share by between 5.5 and 8.8 percentage points, or 13% to 20%, of the average in the sample (including sole led loans). Columns 2 through 4 show consistent results after including lender and borrower fixed effects and loan terms as control variables.

Finally, in panel C, we show that changes in the number of nonbank participants in the next loan increases for disclosers by 0.05 institutions. This effect is quite large given that the average syndicate only has 0.01 institutions. This result implies that disclosure encourages incremental demand for the loan from entities known to trade in public equity markets on information from the loan market (Ivashina and Sun 2011). Then, if it is in fact the borrower's disclosure that engenders the increase in demand for future loans, we would expect entities that utilize this information in other contexts to take advantage of it directly in the loan market as well. Our results suggest that this is indeed the case, providing further support for borrower information sharing as the underlying mechanism. When adding borrower and lender fixed effects, the marginal increase in the institutional share for disclosing borrowers is even larger.

2.3 Loan terms

The results presented in the previous subsections show that borrowers' voluntary information sharing leads to increases in the incidence of lender switching and a broadening of lending syndicates. We would like to infer from these findings that voluntary disclosure by the firm serves to increase lender competition. The missing piece in this argument is whether or not borrowers actually materially benefit from this switching in future loans (Petersen and Rajan 1994). Were this not the case, it could be that the public information sharing somehow hurts the relationship between the borrower and the existing lender, perhaps rendering both parties worse off.

Table 4
Voluntary disclosure and future loan terms

A. Spreads				
	(1)	(2)	(3)	(4)
$Disclose_{it}$	-3.4825** (1.4482)	-3.9445** (1.4473)	-3.6561** (1.5057)	-4.0245** (1.5404)
Loan controls	No	No	No	Yes
Fixed effects:				
Year	Yes	Yes	Yes	Yes
Borrower	No	Yes	Yes	Yes
Lender	No	No	Yes	Yes
R^2	.0200	.0232	.0348	.0383
Obs.	30,628			

B. Amounts				
	(1)	(2)	(3)	(4)
$Disclose_{it}$	8.0596** (3.0682)	7.9003** (2.9639)	6.4614** (2.8564)	8.0362*** (2.8317)
Loan controls	No	No	No	Yes
Fixed effects:				
Year	Yes	Yes	Yes	Yes
Borrower	No	Yes	Yes	Yes
Lender	No	No	Yes	Yes
R^2	.0078	.0100	.0213	.0260
Obs.	30,628			

This table presents fixed effects regression estimates of $\ln Spread_{is}$ and $\ln LoanAmount_{is}$, which are the change in the natural log of the all-in-drawn spreads and loan amounts, respectively, on borrower i 's next (made in period $s > t$) and current loans, on $Disclose_{it}$, an indicator that equals 1 if firm i discloses at least one financial ratio or amount in its year t loan and 0 otherwise, and controls. We include borrower and lender fixed effects to isolate variation in the voluntary disclosure choice that varies across borrowers for the same lender and across loans for the same borrower, and we include year fixed effects to eliminate time-varying economic conditions that might affect voluntary disclosure choices and loan terms. Controls include *SoleLender*, an indicator that identifies loans with a single lender, the money terms of the loan, including $\ln Spread$, $\ln FacilityAmount$, and *Maturity*, and *NumCovenants*, the number of financial covenants in the loan package. $\ln Spread$ and $\ln LoanAmount$ are excluded as controls in regressions in which they are used as dependent variables. Robust standard errors are clustered by borrower and reported in parentheses. * $p < .1$; ** $p < .05$; *** $p < .01$.

In Table 4, we present results of estimating Equation (1) to study the effects of disclosure on subsequent spreads (panel A) and loan amounts (panel B). In these tests, we require current and future spreads and amounts to be observed. In panel A, Column 1 shows that disclosers benefit from 3.5% lower spreads in their next loan. This improvement in terms is accompanied, not surprisingly, by larger loan amounts, as shown in Column 1 of panel B, as these increase by 8.1% after disclosure. The fact that borrowers get better future loan terms provides strong evidence that the initial disclosure was beneficial to the borrower, that is, holdup was mitigated (Schenone 2010).

After the inclusion of borrower and lender fixed effects, we continue to find statistically significantly lower spreads and higher loan amounts for disclosing borrowers. Interestingly, though, while the magnitude of the spread effect increases slightly to a 3.7% reduction, the magnitude of the loan amount effect modestly decreases to 6.5%. This likely reflects a trade-off between spreads and loan amounts, and highlights the benefits of information sharing, since disclosing borrowers actually benefit along both margins.

In untabulated tests, we also experiment with interacting the effect of borrower information sharing with a dummy variable indicating a new lender to investigate whether actually switching lenders is necessary to enjoy the benefits of higher bargaining power. For both outcomes, we find that the interaction coefficients are not statistically significant. This means that disclosers are able to achieve the bargaining position improvement due to their disclosure without actually switching lenders; an improved outside option seems to be sufficient (Sharpe 1990; Von Thadden 2004). These results provide evidence that voluntary information sharing can be a valuable mechanism that mitigates information asymmetries and reduces financial constraints for private firms, allowing them to borrow larger amounts at better terms.

The lender may potentially want these disclosures included because of contractibility or verifiability gained by including these in legal documents. However, if they are explicitly contracted on, this would be measured as a covenant, rather than a voluntary disclosure in our setting. Nonetheless, in all of our tests, we control for covenant use in case there is some complementarity between voluntary disclosure and indirect disclosure through covenants. Even beyond simply controlling for covenants and eliminating disclosures about ratios already contracted upon, the fact that loan terms improve from disclosure is inconsistent with a purely covenant-based story of disclosure. One might wonder whether disclosure benefits borrowers directly by providing benefits on the same loan from which the disclosure takes place. However, this is unlikely given that the lender likely already has significant nonpublic information about the borrower, gleaned from the process of due diligence. Moreover, if lenders play an important role in the disclosure decision, we would expect to observe differences in disclosure across lenders. In an untabulated test, we investigate the importance of lender fixed effects in explaining variation in disclosure and find that it has limited incremental explanatory power beyond borrower and year fixed effects, consistent with the notion that lenders do not drive disclosure choices.

3. The Borrower's Information Sharing Trade-off

The significant benefits of information sharing documented above lead to the obvious question of why only a minority of borrowers choose to disclose. In this section, we investigate the trade-off faced by borrowers in making the information sharing decision and how this affects the interpretation of our empirical evidence. It is possible that significant heterogeneity in borrower benefits means that disclosing borrowers happened to face a particularly large marginal benefit of disclosing, whereas nondisclosing borrowers chose not to disclose because their marginal benefits were small or nonexistent. More plausibly, borrowers face some countervailing costs of disclosure. For example, by disclosing some financial information in their loan documents, borrowers potentially open themselves up to litigation. However, this consequence of

disclosure also strengthens the credibility of the information, thus reinforcing the enhanced lender competition channel. A likely more important cost of disclosing relates to product market competition. The following subsection investigates the determinants of the borrowers' disclosure choice. Our overall goal is to both better understand the economics of borrower information sharing and pin down the nature of the bias underlying our results on the financing benefits of disclosure.

3.1 The cost of sharing information

In the public firm setting, proprietary costs have been shown to be an empirically significant driver of nondisclosure (e.g., Hayes and Lundholm 1996).¹³ Theoretically, it would make sense that proprietary costs also would be an important consideration for private firms, since they too must worry about product market competition. In fact, one might even expect that in the absence of mandatory disclosure of detailed financial information, a marginal disclosure by a private firm would be relatively more informative to its competitors than would be the case for a public firm. We investigate cross-sectional variables that are potentially related to the cost-benefit tradeoff underlying the disclosure choice, as follows:

$$Disclose_{it} = \beta CostBenefit_{it} + \tau X_{it} + \xi_{it}, \quad (2)$$

where $Disclose_{it}$ is an indicator that equals 1 if firm i discloses at least one financial ratio or amount in its year t loan, $CostBenefit_{it}$ is one of several measures of the costs or benefits of disclosure and X_{it} is a set of loan controls, like in Equation (1). We consider an estimation sample for these tests that includes loans only if all variables of interest are nonmissing. Note that most of the determinants we consider are effectively measured at the industry or industry-year level, both because of the fundamental nature of interfirm competition and because we have only limited borrower-specific information for our sample of private firms. For that reason, we do not include other fixed effects, such as at the borrower level, in our specifications.

Table 5 presents results on the relationship between voluntary borrower information sharing and drivers of disclosure drawn from the literature that typically studies public firms. First, Column 1 looks at $R\&DIntensity$, which is defined as the average ratio of research and development (R&D) expenses to total assets among public companies in borrower i 's two-digit SIC industry in year t . We find that greater $R\&DIntensity$ is associated with less borrower information sharing, consistent with the idea that more innovative firms that depend on research and development more are less likely to disclose because proprietary costs are higher (Jones 2007; Ellis, Fee, and Thomas 2012).

¹³ Other costs (and benefits) that we do not study are likely relevant to the disclosure decision. Our goal in this subsection is to highlight the role, in the *private* firm setting, of typical drivers of disclosure choice in the *public* firm setting.

Table 5
The costs of voluntary disclosure

	<i>Disclose_{it}</i>							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>RandDIntensity_{it}</i>	-13.5535** (5.8833)							
<i>HHI_{it}</i>		-10.7481* (6.3388)						
<i>NumCompetitors_{it}</i>			0.0697*** (0.0127)					
<i>MarginPersistence_{it}</i>				-27.5106*** (2.3615)				
<i>ProfitMargin_{it}</i>					0.2495*** (0.0561)			
<i>AverageGrowth_{it}</i>						0.7851*** (0.2283)		
<i>GrowthDispersion_{it}</i>							0.1913** (0.0839)	
<i>LitigationRisk_i</i>								-6.3247*** (1.1901)
Loan controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
<i>R</i> ²	.0532	.0501	.0559	.0646	.0488	.0490	.0526	.0560
Obs.	1,800							

This table presents fixed effects regression estimates of *Disclose_{it}*, an indicator that equals 1 if firm *i* discloses at least one financial ratio or amount in its year *t* loan and 0 otherwise, on measures of the innovative, proprietary, competitive, and legal costs of disclosure plus controls. *RandDIntensity* measures the average ratio of RandD expense to total assets among public companies in borrower *i*'s two-digit SIC industry in year *t*. *HHI* is the Hirschman-Herfindahl index measure of concentration based on the revenues of public companies in borrower *i*'s two-digit SIC industry and state in year *t*. *NumCompetitors* is the count of public companies in borrower *i*'s two-digit SIC industry and state in year *t*. *MarginPersistence* measures the autocorrelation in abnormal (i.e., industry-adjusted) profits of public companies in borrower *i*'s two-digit SIC industry. *ProfitMargin* is the average ratio of net income to revenue of public companies in borrower *i*'s two-digit SIC industry and state in year *t*. *AverageGrowth* is the average sales growth of public companies in borrower *i*'s two-digit SIC industry and state in year *t*. *GrowthDispersion* is the standard deviation of sales growth for public companies in borrower *i*'s two-digit SIC industry and state in year *t*. *LitigationRisk* is an indicator that equals 1 if borrower *i*'s primary SIC industry is a high litigation risk industry (Kim and Skinner 2012). Controls include *SoleLender*, an indicator that identifies loans with a single lender, the money terms of the loan, including *lnSpread*, *lnFacilityAmount*, and *Maturity*, and *NumCovenants*, the number of financial covenants in the loan package. The sample is restricted to loans for which all regression variables of interest are nonmissing. Robust standard errors are clustered by borrower and reported in parentheses. **p* < .1; ***p* < .05; ****p* < .01.

In Column 2, we investigate competition using the Hirschman-Herfindahl index (Herfindahl 1950; Hirschman 1964; Giroud and Mueller 2011), measured like above using public companies, and find that borrowers are less likely to share information when an industry is more concentrated (Bamber and Cheon 1998). This finding, in concert with the positive correlation between the number of industry competitors and the likelihood of disclosure seen in Column 3, shows that firms with more monopoly power are more concerned with sharing information with their actual or potential competitors.

To provide evidence that the cross-sectional variation in disclosure is indeed driven by differences in the value, to competitors, of the information that would be disclosed, we next investigate several measures of profitability and growth. *MarginPersistence* is the autocorrelation in abnormal (i.e., industry-adjusted) profits of public companies in the borrower's two-digit SIC industry and is winsorized at the 5% level. *ProfitMargin* is the average ratio of net income to

revenue of public companies in borrower i 's two-digit SIC industry and state in year t . *AverageGrowth* is the average sales growth of public companies in borrower i 's two-digit SIC industry and state in year t . *GrowthDispersion* is the standard deviation of sales growth for public companies in borrower i 's two-digit SIC industry and state in year t .

We find that borrower information sharing is less likely when the borrower's industry features stable profit margins, but more likely when profit margins are high. If profit margins are stable in an industry, then information is less noisy, and so, more useful to competitors. Firms in these industries may have more to lose from disclosure; however, firms in high performing industries may have more to gain because their information is more likely to be positive. In fact, higher growth industries see more borrower information sharing, as do industries where there is greater sales growth dispersion. These last two findings suggest that information sharing is more likely when the information is more likely to be positive, and when there is more opportunity for a firm to distinguish itself from its competitors, like in Verrecchia (1983). These results highlight a trade-off in making the information sharing decision, where both the costs and benefits of disclosure are higher for "better" news and suggest that the direction of any empirical bias from unobservable firm quality is ambiguous. We return to this point in Section 3.3.

Finally, we investigate the litigation risk story mentioned above, where we measure *LitigationRisk* using an indicator equal to 1 if the borrower's primary SIC industry is a high litigation risk industry, like in Kim and Skinner (2012). Consistent with our story, high litigation risk is associated with less borrower information sharing, suggesting that, all else equal, the cost of potential litigation outweighs the benefits of "hardening" the information disclosed to other potential lenders. Overall, the results in Table 5 show that the typical drivers of the disclosure choice in the public firm setting also have significant explanatory power for the information sharing decisions of private borrowers.

3.2 The IPO setting

The subsample of private firms that eventually do an IPO provides an ideal laboratory in which to study the trade-off underlying the borrower information sharing decision. As the firm gets close to its IPO, which is preceded by very broad mandatory disclosure requirements, the cost of disclosure falls, as the revelation of proprietary information to competitors becomes inevitable. In advance of a big financing event, such as an IPO, the marginal benefits of disclosure, through the channel of driving supply of potential sources of finance, may also increase.¹⁴ Thus we expect that borrower information sharing should unambiguously increase as the IPO gets closer. Unlike the proxies for the cost of information sharing discussed in Section 3.1, the IPO setting

¹⁴ Consistent with this idea, in Section 3.4, we show that borrower information sharing does in fact lead to improved IPO outcomes.

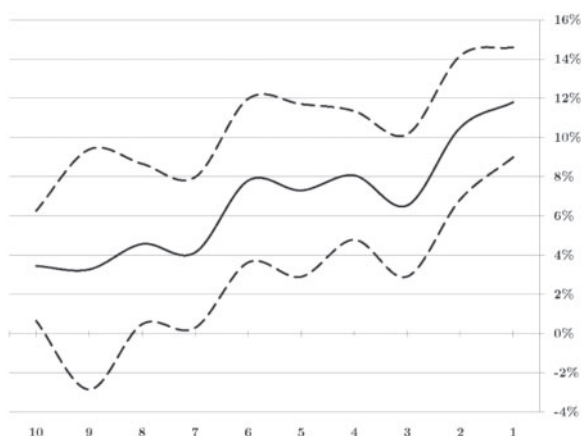


Figure 1
Probability of disclosure and time to IPO

This figure plots the propensity of loans to include disclosures of financial ratios and amounts against the number of years before the borrower makes an initial public offering (IPO). The solid line represents the 3-year smoothed average probability of disclosure, and the dashed lines represent the 90% confidence interval around the average probability of disclosure for each year prior to IPO.

provides within-firm variation in the net benefits of sharing information, which allows for more discriminating empirical tests. Figure 1 shows the relationship between disclosure and years to IPO. The likelihood of borrower information sharing increases leading up to an IPO. However, this relationship could be due to the changes in the characteristics of loans or secular macroeconomic factors, and so we investigate the disclosure trend in a regression framework.

Table 6 presents results from estimating Equation (2) after including time to IPO on the right-hand side.¹⁵ In Columns 1 and 2, we estimate the effects of time to IPO nonparametrically, whereas in Columns 3 and 4, we adopt a linear specification. Column 1 shows that borrowers are 3.6 percentage points more likely to disclose sales or key financial ratios in the 2 years prior to their IPO versus 2.7 percentage points more likely to disclose in the next 3 years prior. These estimates correspond to 30% to 40% increases relative to the mean probability of disclosure in this sample of 9%. In the next column, with disclosure modeled as a linear function of time to IPO, we expect the preceding results to manifest themselves in a negative coefficient. That is indeed what we find, as the probability of disclosure increases by 0.3 percentage points per year closer to the IPO.

¹⁵ The substantially smaller sample of loans from private firms that eventually raise public financing renders the inclusion of borrower and lender fixed effects impractical. This is a trade-off of using the IPO sample.

Table 6
Voluntary disclosure before initial public offerings

	<i>Disclose_{it}</i>			
	Nonparametric		OLS	
	(1)	(2)	(3)	(4)
1[<i>TimetoIPO</i> ≤ 2]	3.6393** (1.4586)	3.1944** (1.2385)		
1[3 ≤ <i>TimetoIPO</i> < 6]	2.6845** (1.1752)	1.9675* (1.1407)		
<i>TimetoIPO_{it}</i>			−0.4714** (0.2079)	−0.3425* (0.1877)
Loan controls	No	Yes	No	Yes
Fixed effects:				
Year	Yes	Yes	Yes	Yes
<i>R</i> ²	.0774	.1415	.0771	.1407
Obs.	1,683			

This table presents fixed effects regression estimates of *Disclose_{it}*, an indicator that equals 1 if firm *i* discloses at least one financial ratio or amount in its year *t* loan and 0 otherwise, on measures of the amount of time until the borrower makes its first public equity offering plus controls. *TimetoIPO* is the natural log of the number of years before the firm makes a public equity issuance. The corresponding indicator variables in Column 1 investigate the nonparametrical relationship between *TimetoIPO* and *Disclose* by identifying observations from the 2 years before borrower *i*'s initial public offering and observations from 3 and 4 years before borrower *i*'s initial public offering. Controls include *SoleLender*, an indicator that identifies loans with a single lender, the money terms of the loan, including *LnSpread*, *LnFacilityAmount*, and *Maturity*, and *NumCovenants*, the number of financial covenants in the loan package. Robust standard errors are clustered by borrower and reported in parentheses. **p* < .1; ***p* < .05; ****p* < .01.

3.3 Unobservable borrower quality

The main empirical concern with interpreting the results in Section 2 is the potential for bias from unobserved borrower quality. Specifically, if better borrowers are more likely to disclose and also achieve better financing outcomes, then this channel could provide an alternative explanation. The control variables and fixed effects in our earlier specifications should help address this issue, but may not be sufficient. To start, we consider potential theoretical mechanisms that could tie together disclosure and borrower quality.

The theoretical intuition of Verrecchia (1983) provides the foundation for such a link. Higher-quality borrowers are more likely to disclose in order to reveal their type and so avoid pooling with nondisclosing borrowers. However, several complicating factors push in the opposite direction. For example, we showed above that proprietary costs discourage disclosure. Proprietary costs should be higher for better quality firms, and since the relatively positive signal would be more likely to induce competition, this mechanism would make them less likely to disclose.

In a richer model, the relationship between the benefit of disclosure and firm quality is also less clear. All else equal, lenders are interested in lending to high quality firms, so these borrowers receive plenty of attention. However, lower-quality firms may need to take action to attract such interest. Publicly sharing key financial information is a means of accomplishing this “marketing” goal. A distinguishing feature of the private firm setting is that the meaning

of disclosure and the importance and nature of news is different. When the information environment of a firm is rich, incremental disclosure will be benchmarked against existing information. This leads to a trade-off in disclosure models between separating by disclosing good information and pooling with all nondisclosing firms. However, it is less meaningful to think about information, like sales, as “good” or “bad” when no obvious benchmark exists. More generally, the bank will not compete for a borrower about which it is unaware or has only little or imprecise information. For these reasons, the difference in average unobservable quality of disclosers and nondisclosers is theoretically ambiguous and so must be investigated empirically.

While the fixed effects included in the main specification suggest that the bias from unobserved quality may not meaningfully affect our inferences, there is still a possibility that time-varying borrower quality could cause problems. The IPO sample introduced above provides an opportunity to investigate the bias, particularly because the life cycle of the firm, measured by its proximity to IPO, is a good predictor of the information sharing choice. Further, going public is a key milestone in the life cycle of a firm in which the firm will reveal its quality to a broader market of investors, so we may infer that firms pursuing an IPO select into that decision based on unobservable characteristics. Our first variable, $TimetoIPO_{it}$, controls for changes in firm characteristics as they approach the IPO; this may be related to quality since firms strategically choose when to IPO. Our second variable, T_i^{MAX} , which is the total time spent as a private firm, controls for fixed characteristics of a firm that make remaining private beneficial. On average, we expect that a firm that is quick to IPO is likely to have unobservably good growth opportunities.¹⁶ We start by simply including these firm-specific timing variables as controls to get the following estimating equation:

$$Y_{is} = \theta Disclose_{it} + \gamma T_i^{MAX} + \psi TimetoIPO_{it} + \rho X_{it} + v_t + \xi_{it}. \quad (3)$$

Next, we draw on an IV strategy from the labor economics literature as an alternative way to use this life-cycle variation to combat endogenous borrower quality. Specifically, we use an estimator developed in Altonji and Shakotko (1987) to study the impact of employee tenure on wages. In that context, employee tenure is likely related to unobservable differences in employee quality or unobservable differences in the quality of the match between employee and employer. In our setting, we are interested in controlling for unobservable differences in firm quality or unobservable differences in the quality of the match between the firm and its private status, as proxied by the total number of years the firm exists in this data set before its IPO.

¹⁶ Of course, exceptions to this logic exist. Goldman Sachs, for example, was private for 130 years before going public in 1999. Note that the year fixed effects included in all specifications relate to the year of the loan; however, our results are qualitatively robust to including year-of-IPO fixed effects to address macroeconomic or regulatory shocks to IPO decisions.

We provide a simple numerical explanation of the Altonji and Shakotko (1987) transformation in Appendix A.

This strategy relies on variation in the number of years until initial public equity issuance, adjusted for the total number of years between first private loan initiation and first public issuance. That is, the estimator transforms the number of years until initial public issuance into the fraction of time until initial public issuance that has passed. Then, information sharing that occurs 1 year prior to public issuance for a firm that goes public 5 years after its first loan, and information sharing that occurs 2 years prior to public issuance for a firm that goes public 10 years after its first loan are treated similarly. Both disclosures occur 80% of the way through their respective firms' pre-IPO life, which removes the effects of unobservable differences between these firms that result in their decisions to remain private for 5 and 10 years, respectively, after their first loan. This strategy collapses time to IPO into the same unit interval for all firms, effectively creating a quality-adjusted disclosure trend for each firm. The reason this method works is that it effectively only uses variation from the loan being the first loan, a middle loan, or the last loan before an IPO, and it is difficult to argue that all firms get better in unobservable ways from first to last loan.

We implement the Altonji and Shakotko (1987) method with the following three stage model:

$$\begin{aligned} \text{Time to IPO}_{it} &= \delta_0 + \delta_1 \frac{t - T_i^0}{T_i^{\text{MAX}}} + \kappa X_{it} + v_t + e_{it}, \\ \widehat{\text{Disclose}}_{it} &= \widehat{\text{Time to IPO}}_{it} + \tau X_{it} + v_t + \varepsilon_{it}, \\ Y_{is} &= \theta \widehat{\text{Disclose}}_{it} + \rho X_{it} + v_t + \xi_{it}, \end{aligned} \quad (4)$$

where T_i^0 is the first year the firm appears in our sample, such that the instrument for time to IPO is the fraction of time elapsed between when the firm first appeared in the sample and when it does its IPO. We then use the predicted $\widehat{\text{Disclose}}$ as an instrument for a loan outcome in some future period s .

Table 7 shows the estimates of borrower information sharing on the incidence of lender switching for the IPO sample. This table corresponds to Table 2, with the smaller sample being the primary difference. Again, we find that borrower information sharing is associated with more lender switching on subsequent loans, implying that the mechanism of enhanced lender competition applies similarly in the subsample of private firms which eventually IPO. In Column 2, we add the two timing variables as controls, and we estimate a 15.0 percentage point effect of disclosure on lender switching, which is similar to our estimates from Table 2 and those in Column 1 of this table. Finally, the magnitude of our estimates from the IV model in Equation (4) is substantially larger. This increase could be due, in part, to estimation noise in our three-stage empirical model akin to a weak instruments problem. However, when combined with

Table 7
Voluntary disclosure and lender choice: Instrumental variables

	<i>NewLender_{is}</i>		
	OLS	OLS	IV
	(1)	(2)	(3)
<i>Disclose_{it}</i>	16.6520*** (4.9814)	14.9983*** (4.8528)	53.1173* (31.3892)
Loan controls	Yes	Yes	Yes
Timing controls	No	Yes	No
Fixed effects:			
Year	Yes	Yes	Yes
<i>R</i> ²	.2681	.2852	
Obs.	1,186		

This table presents fixed effects and instrumental variables regression estimates of measures of lender switching on *Disclose_{it}*, an indicator that equals 1 if firm *i* discloses at least one financial ratio or amount in its year *t* loan and 0 otherwise, and controls. *NewLender_{is}* is an indicator that equals 1 if the borrower chooses a new lender on its subsequent loan (in period *s* > *t*). Column 1 presents an OLS model; Column 2 presents an OLS model with timing controls, *TimetoIPO* and *T^{Max}*, the total number of years that borrower *i* spent private, as controls for selection on disclosure timing and borrowers' endogenous match quality with private status; and Column 3 presents an instrumental variables model following Altonji and Shakotko (1987), as discussed in detail in Section 3.3 and Appendix A. Loan controls include *SoleLender*, an indicator that identifies loans with a single lender, the money terms of the loan, including *InSpread*, *InFacilityAmount*, and *Maturity*, and *NumCovenants*, the number of financial covenants in the loan package. Robust standard errors are clustered by borrower and reported in parentheses in Columns 1 and 2, and bootstrapped robust standard errors are presented in Column 3. **p* < .1; ***p* < .05; ****p* < .01.

our fixed effects evidence from earlier tables, the increasing magnitude also suggests that unobserved borrower quality has an attenuating effect. The fact that both strategies that use time to IPO yield statistically and economically significant results is quite interesting, in that one approach effectively throws out time series variation, while the other exploits it, suggesting that very different sources of variation are pushing in the same direction. We find a similar pattern of results when we study the effect of information sharing on syndicate structure in Table 8. Like above, adding timing controls or following the IV strategy both yield statistically and economically significant estimates.

Finally, Table 9 shows the effect of voluntary information sharing on changes in loan spreads and amounts. For both financing outcomes, the addition of the timing controls does not much change the magnitude of the effect of interest, while the IV strategy again yields larger magnitudes in the case of spreads. Overall, this additional set of results does not change our inferences regarding the mechanism whereby borrower information sharing enhances lender competition, which benefits the borrower in terms of both lower spreads and larger amounts. The consistent theme of the results is that across a variety of strategies for dealing with unobserved borrower heterogeneity, the estimated benefits of disclosure are typically higher than in our baseline specification. This suggests that, if anything, lower-quality borrowers are more likely to disclose. We attribute this finding to the fact that worse-off firms have more to gain from a marginal increase in lender competition, because good firms already attract enough attention to engender competition.

Table 8
Voluntary disclosure and next syndicate structure: instrumental variables

	NumLenders _{is}			RetainedShare _{is}			NumInstitutions _{is}		
	OLS	OLS	IV	OLS	OLS	IV	OLS	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Disclose _{it}	0.5628** (0.2744)	0.5237** (0.2619)	2.3642* (1.3968)	-10.3931* (5.6720)	-10.9086* (5.7751)	-24.2894** (10.5722)	0.0572* (0.0329)	0.0569* (0.0326)	0.3011*** (0.1011)
Loan controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Timing controls	No	Yes	No	No	Yes	No	No	Yes	No
Fixed effects:									
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
R ²	.1017	.1092		.7680	.7708		.0270	.0277	
Obs.	1,683								

This table presents fixed effects and instrumental variables regression estimates of $NumLenders_{is}$, the difference between the number of lenders participating in borrower i 's next and current loans, $RetainedShare_{is}$, the difference between the lead bank's retained share on borrower i 's next and current loans, and $NumInstitutions_{is}$, the difference between the number of nonbank participants in borrower i 's next and current loans, on $Disclose_{it}$, an indicator that equals 1 if firm i discloses at least one financial ratio or amount in its year t loan and 0 otherwise, and controls. Columns 1, 4, and 7 present OLS models; Columns 2, 5, and 8 present OLS models with timing controls, $TimetoIPO$ and T^{Max} , the total number of years that borrower i spent private, as controls for selection on disclosure timing and borrowers' endogenous match quality with private status; and Columns 3, 6, and 9 present instrumental variables models following Altonji and Shakotko (1987), as discussed in detail in Section 3.3 and Appendix A. Loan controls include $SoleLender$, an indicator that identifies loans with a single lender, the money terms of the loan, including $lnSpread$, $lnFacilityAmount$, and $Maturity$, and $NumCovenants$, the number of financial covenants in the loan package. Robust standard errors are clustered by borrower and reported in parentheses in Columns 1, 2, 4, 5, 7, and 8, and bootstrapped robust standard errors are presented in Columns 3, 6, and 9. * $p < .1$; ** $p < .05$; *** $p < .01$.

Table 9
Voluntary disclosure and future loan terms: Instrumental variables

	lnSpread _{is}			lnLoanAmount _{is}		
	OLS	OLS	IV	OLS	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Disclose _{it}	-4.8296** (2.3179)	-5.0279** (2.4037)	-25.1832** (10.1343)	21.8325** (8.9412)	22.2876** (9.2197)	22.1786*** (6.9604)
Loan controls	Yes	Yes	Yes	Yes	Yes	Yes
Timing controls	No	Yes	No	No	Yes	No
Fixed effects:						
Year	Yes	Yes	Yes	Yes	Yes	Yes
R ²	.0249	.0257		.1251	.1284	
Obs.	1,683					

This table presents fixed effects and instrumental variables regression estimates of $lnSpread_{is}$ and $lnLoanAmount_{is}$, which are the change in the natural log of the all-in-drawn spreads and loan amounts, respectively, on borrower i 's next (made in period $s > t$) and current loans, on $Disclose_{it}$, an indicator that equals 1 if firm i discloses at least 1 financial ratio or amount in its year t loan and 0 otherwise, and controls. Columns 1 and 4 present OLS models; columns 2 and 5 present OLS models with timing controls, $TimetoIPO$ and T^{Max} , the total number of years that borrower i spent private, as controls for selection on disclosure timing and borrowers' endogenous match quality with private status; and Columns 3 and 6 present instrumental variables models following Altonji and Shakotko (1987), as discussed in detail in Section 3.3 and Appendix A. Controls include $SoleLender$, an indicator that identifies loans with a single lender, the money terms of the loan, including $lnSpread$, $lnFacilityAmount$, and $Maturity$, and $NumCovenants$, the number of financial covenants in the loan package. $lnSpread$ and $lnLoanAmount$ are excluded as controls in regressions in which they are used as dependent variables. Robust standard errors are clustered by borrower and reported in parentheses in Columns 1, 2, 4, and 5, and bootstrapped robust standard errors are presented in Columns 3 and 6. * $p < .1$; ** $p < .05$; *** $p < .01$.

While no individual specification is perfect, we believe that our multipronged empirical strategy, including the use of borrower and lender fixed effects in our main tests, multiple samples, and several uses of firm life-cycle variables based on time to IPO, provides strong evidence that it is indeed the voluntary information sharing by the borrower, rather than unobservable differences between disclosers and nondisclosers, that explains the increase in lender competition and improvement in financing outcomes.

3.4 Information sharing and public offering outcomes

To complete our analysis of the IPO sample, our last set of tests relate to the effects of borrower information sharing on the public financing event itself. The effects of disclosure on both existing and future lending relationships suggest the possibility of similar effects on IPO outcomes. Disclosing could also improve the borrower’s bargaining power during the process of finding and negotiating with underwriters and so may result in better outcomes for the firm, which is related to the “tying” between lenders and underwriters documented in Drucker and Puri (2005). Basically, if the lender has an information advantage in the private lending market, we might expect this information advantage to manifest itself in the underwriting process as well.

Econometrically, the IPO setting has an additional advantage relative to modeling loan outcomes in that it is a one-time event. This means that any results are unlikely to be explained by time-varying firm characteristics. As such, the results we find for IPO outcomes support our inferences about the effect of borrower information sharing on lending competition. Table 10 presents

Table 10
Voluntary disclosure and initial public offerings

	lnIPOAmount _i			Underpricing _i		
	OLS	OLS	IV	OLS	OLS	IV
	(1)	(2)	(3)	(4)	(5)	(6)
Disclose _{it}	0.3575*** (0.1226)	0.3638*** (0.1129)	3.3019*** (0.6462)	−0.1838*** (0.0691)	−0.1876*** (0.0669)	−0.8619** (0.4184)
Loan controls	Yes	Yes	Yes	Yes	Yes	Yes
Timing controls	No	Yes	No	No	Yes	No
Fixed effects:						
Year	Yes	Yes	Yes	Yes	Yes	Yes
R ²	.2515	.2721		.1450	.1490	
Obs.	1.167					

This table presents fixed effects and instrumental variables regression estimates of lnIPOAmount_i, the natural log of the dollar amount of borrower *i*’s initial public offering and Underpricing_i, the underpricing percentage value of firm *i*’s IPO, on Disclose_{it}, an indicator that equals 1 if firm *i* discloses at least one financial ratio or amount in its year *t* loan and 0 otherwise, and controls. Columns 1 and 4 present OLS models; Columns 2 and 5 present OLS models with timing controls, *TimetoIPO* and *T^{Max}*, the total number of years that borrower *i* spent private, as controls for selection on disclosure timing and borrowers’ endogenous match quality with private status; and Columns 3 and 6 present instrumental variables models following Altonji and Shakotko (1987), as discussed in detail in Section 3.3 and Appendix A. Loan controls include *SoleLender*, an indicator that identifies loans with a single lender, the money terms of the loan, including lnSpread, lnFacilityAmount, and Maturity, and NumCovenants, the number of financial covenants in the loan package. Robust standard errors are clustered by borrower and reported in parentheses in Columns 1, 2, 4, and 5, and bootstrapped robust standard errors are presented in Columns 3 and 6. **p* < .1; ***p* < .05; ****p* < .01.

our results on the effects of voluntary disclosure on the characteristics of the subsequent IPO, and we restrict the estimation sample for these tests to include observations with nonmissing values for offer amount and underpricing. Columns 1 to 3 present estimates of effect of information sharing on IPO amounts, the idea being that firms with more bargaining power will be able to raise more funds during their IPOs, either because of better pricing from underwriters or greater demand from investors due to reduced information asymmetry. The IV estimate in Column 3 suggests that disclosure results in a 3.3% increase in the subsequent IPO amount, an effect that is statistically significant at the 1% level.

We look at underpricing in Columns 4 to 6 since this is a direct measure of the bargaining power of the firm relative to the underwriters (Ljungqvist 2005), and we find a significant reduction in underpricing. Across specifications, the magnitude varies between 18 and 86 bps, or between 2.5% and 12.3%, of the average level of underpricing in our sample. Controlling for the timing variables, we obtain results that are similar in magnitude and statistical significance. This is not surprising, given that the IPO is fixed in time relative to the time to IPO-driven trend in disclosure. Our IV estimates are somewhat larger in magnitude, which could be due either to more effective control for unobservable borrower quality or measurement error in the three-stage empirical model. Overall, the results in Table 10 reveal that public information sharing allows firms better access to financing, not just in private lending relationships but also from public markets. This suggests that the disclosure can mitigate asymmetric information between the borrower and providers of public equity.

4. Conclusion

The information advantage of incumbent lenders relative to prospective outside lenders is likely to be particularly large for private borrowers, given the paucity of required mandatory public disclosure. To mitigate the holdup problem induced by this information asymmetry, borrowers can publicly share information with these potential lenders through the disclosure of financials in loan documents, which are collected, aggregated, and made available to all lenders by LPC. We find that borrowers who choose to share information are more likely to switch lenders and borrow from larger syndicates. This increased supply of credit leads to lower spreads and larger loan amounts for disclosing borrowers. Our results are robust to a variety of ways of dealing with the issue of unobserved borrower quality, including an IV strategy based on the life cycle of private firms that eventually go public. In fact, we find that borrowers with unobservable low-quality characteristics are more likely to benefit from sharing information with prospective lenders. In making their information sharing decision, borrowers appear to trade off the financing benefits against the cost of providing valuable information to product market competitors. Overall,

we find that voluntary borrower information sharing reduces lender holdup by enhancing lender competition, and thus is an important mechanism for alleviating financial constraints for private firms.

Appendix A

What follows is a stylized version of the actual two-stage least squares model we used to develop intuition:

$$\text{First stage: } T = \alpha^1 + \beta^1 \left(\frac{T - \bar{T}}{T_{\max}} \right) + \epsilon^1, \quad (\text{A1})$$

$$\text{Second stage: } \text{Disclose} = \alpha^2 + \beta^2 \hat{T} + \epsilon^2. \quad (\text{A2})$$

The first-stage model instruments for T , the time until first public issuance, using $\left(\frac{T - \bar{T}}{T_{\max}} \right)$, or the time until public issuance minus the average time until public issuance, divided by the total number of years between first loan and initial public issuance. This construction, first analyzed by Altonji and Shakotko (1987), removes variation in T , or time to first public issuance, that comes from $T_{\max}$. It is ideal in the sense that it satisfies the exclusion restriction by construction. Specifically, like in Altonji and Shakotko (1987), $E \left[\left(\frac{T - \bar{T}}{T_{\max}} \right) \times \epsilon^1 \right] = 0$ by the properties of expectations. Similarly, $\frac{T - \bar{T}}{T_{\max}}$ satisfies the relevance criteria because it is positively correlated with T by the properties of covariance (Altonji and Shakotko 1987). The second stage identifies the coefficient β , the effect of time until initial public issuance on voluntary disclosure, using the quality-adjusted variation (i.e., the fraction of time until initial public issuance that has elapsed).

A.1 A Simple Example of Quality-Adjusted Trends in Disclosure

Next, we present a simple example to illustrate how the instrument transforms the time to initial public issuance for firms and why we expect the endogeneity from unobservable differences in firm quality and firm-private status match quality to be attenuating. The purpose of this example is to demonstrate how a firm with shorter time to IPO has its loan disclosures stretched out over time, whereas a firm with longer time to IPO has its loan disclosures compressed, placing trend-driven disclosure on the same scale as a way of removing the influence of time-invariant differences across firms.

Suppose firm 1 has $T_{\max} = 2$ and firm 2 has $T_{\max} = 6$. The time line of each firms' years until IPO is as follows:

Firm 1: 2, 1, 0,

Firm 2: 6, 5, 4, 3, 2, 1, 0.

Then the instrument for years to IPO is as follows:

Firm 1 (instrumented): 1/2, 0, -1/2,

Firm 2 (instrumented): 1/2, 1/3, 1/6, 0, -1/6, -1/3, -1/2.

This means that any loan disclosures made by firm 1 will have an increased impact on instrumented time to IPO coefficients, whereas the effects for firm 2 will be lower. Therefore, the use of time to IPO controls for the effects of unobservable firm quality on voluntary financial disclosure in private loans.

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